

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.025
##
## $beta.var.scale
## [1] 0.3
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.3992  0.4325  0.4433  0.4427  0.4525  0.4892

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0925  0.1408  0.1372  0.1750  0.8975
```

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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.0950  0.1450  0.1400  0.1783  0.7575
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.000000 0.005000 0.008333 0.102119 0.031667 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.000000 0.005000 0.008333 0.077939 0.025000 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.000000 0.005833 0.015000 0.162184 0.118333 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.2158  0.2517  0.2617  0.2610  0.2692  0.3017

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.0500  0.0975  0.1031  0.1383  0.9250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.05167	0.10083	0.10558	0.14333	0.70667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0008333	0.0391667	0.4041622	1.0000000	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.003333	0.241180	0.290000	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08667	1.00000	0.68720	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.06667	0.09333	0.10333	0.10564	0.11417	0.19583

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03104	0.07750	0.08432	0.12083	0.84583

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03333	0.08167	0.08769	0.12667	0.63500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3458	0.5500	0.6008	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.001667	0.261695	0.557083	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5983	1.0000	0.7814	1.0000	1.0000

```

--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.02750 0.04667 0.05333 0.05695 0.06417 0.14667

```

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beta.ar (Location: fit[[4]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01417 0.05250 0.06381 0.09500 0.80250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

```

gamma.ar (Location: fit[[4]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01500 0.05500 0.06712 0.10083 0.68583
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.4525  0.8608  0.6884  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.8892  1.0000  1.0000  1.0000  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2448  0.5025  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.7492  1.0000  0.7954  1.0000  1.0000

```

```

--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.03000 0.05750 0.06750 0.07022 0.07917 0.15417

```

```

beta.ar (Location: fit[[5]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01167 0.04750 0.05910 0.08667 0.80667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01333 0.05167 0.06267 0.09333 0.59917
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.3600  0.4279  0.8358  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.6433  1.0000  1.0000  0.9999  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2087  0.3442  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.7100  0.9208  0.7754  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01250 0.02167 0.02750 0.02845 0.03250 0.08917

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01000 0.03917 0.05395 0.08000 0.79250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01167 0.04333 0.05829 0.08833 0.53250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.3942  0.4566  0.9600  1.0000
  Modality 3 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8208	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2273	0.3840	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7325	0.9967	0.7882	1.0000	1.0000

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01000	0.02583	0.03250	0.03442	0.04000	0.09917

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.035833	0.050844	0.075833	0.758333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.04000	0.05603	0.08500	0.60917

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.2475	0.4058	0.8642	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02833	1.00000	1.00000	0.99990	1.00000	1.00000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1915	0.1165	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5833	0.9083	0.7281	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04167	0.07333	0.08667	0.09178	0.10583	0.20083

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.001667 0.013333 0.031782 0.047500 0.691667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

gamma.ar (Location: fit[[8]]\$gamma.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.001667 0.015000 0.035668 0.055000 0.485000
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1436 0.2158 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.9175 1.0000 0.9485 1.0000 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.06216 0.00000 0.95083
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.2575 0.3156 0.5825 1.0000

```

--- Summary for K = 9 ---

Z.ar (Location: fit[[9]]\$Z.ar):

```

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.1058 0.1325 0.1458 0.1483 0.1600 0.3533

```

beta.ar (Location: fit[[9]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.001667 0.011667 0.024332 0.034167 0.692500
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

```

gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.001667 0.013333 0.027572 0.040000 0.445000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1351 0.1100 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.5150 0.6917 0.6598 0.8408 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01339 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.08974 0.00000 1.00000

```